

# CV-10 Belt Conveyor

Safety, Tracking and Drive Maintenance Reference

**SIMULATED TRAINING DOCUMENT - NOT A MANUFACTURER MANUAL**

## 1. Equipment Identification

|                       |                            |
|-----------------------|----------------------------|
| Equipment code        | CV-10                      |
| Equipment type        | Motor-driven belt conveyor |
| Service               | Packaged material transfer |
| Nominal belt speed    | 0.8 m/s                    |
| Maximum training load | 250 kg                     |
| Tracking tolerance    | +/- 10 mm                  |
| Manual revision       | Training Revision 1.0      |

### System Overview

The CV-10 includes a drive motor, gearbox, head pulley, tail pulley, idlers, tension system, belt, guards and emergency-stop circuit.

#### SAFETY PREREQUISITES

Press emergency stop. Isolate the main electrical disconnect. Apply lockout/tagout. Block stored mechanical energy and verify the belt cannot start. Wear gloves, eye protection and safety footwear. Replace all guards before restart.

## 2. Troubleshooting

### Belt Drifts to One Side

#### SAFETY PREREQUISITES

Press emergency stop. Isolate the main electrical disconnect. Apply lockout/tagout. Block stored mechanical energy and verify the belt cannot start. Wear gloves, eye protection and safety footwear. Replace all guards before restart.

#### Observed symptoms

The belt moves toward one frame rail, edge wear becomes visible, material spills near the loading zone, or an idler produces abnormal noise.

#### Possible causes listed in this manual

Documented causes are unequal idler alignment, material buildup on pulleys, off-center loading, damaged idlers, loose structure, or incorrect belt tension.

#### Inspection guidance

After isolation, clean buildup, inspect idler rotation and alignment, confirm the frame is level, and verify that material enters the belt centrally. Never adjust tracking on a moving belt.

#### Stop / refusal condition

If the safety prerequisites cannot be confirmed, stop. The manual does not authorize diagnosis or repair without isolation, zero-energy verification and required PPE.

### 3. Authorized Inspection Procedure

#### SAFETY PREREQUISITES

Press emergency stop. Isolate the main electrical disconnect. Apply lockout/tagout. Block stored mechanical energy and verify the belt cannot start. Wear gloves, eye protection and safety footwear. Replace all guards before restart.

Step 1. Record the direction and location of belt drift.

Step 2. Stop the conveyor and apply lockout/tagout.

Step 3. Remove material buildup from pulleys and idlers.

Step 4. Inspect idlers for seizure, damage and alignment.

Step 5. Check loading position and belt tension.

Step 6. Make small alignment corrections only while isolated.

Step 7. Replace guards and perform a controlled test under supervision.

#### Reference Parts

|              |                      |
|--------------|----------------------|
| CV10-IDL-20  | Carrying idler       |
| CV10-IDL-21  | Return idler         |
| CV10-BELT-01 | Conveyor belt        |
| CV10-GRD-04  | Drive guard assembly |

**All work orders generated from this document require explicit supervisor review and approval before persistence or execution.**